

Scope Note — Equipment Downtime & Maintenance Cost Analysis

Analyst: Sudhansu Kumar Biswal **Dataset:** Equipment downtime log, fabrication workshop — 686 events, 12 assets, 01-Feb-2025 to 31-Jul-2026 **Document version:** v2 (revised after data review) · September 2026 **Status:** Portfolio project. Dataset is synthetic, generated for demonstration purposes.

1. The problem

A fabrication workshop records every maintenance and breakdown event in a log but has no analysis layer on top of it, so the maintenance manager cannot see where downtime cost is concentrating or decide which assets to prioritise for preventive maintenance.

The reader of this workbook is the plant or maintenance manager. The decision it supports is: **where to spend the next rupee of maintenance effort.**

2. Questions this analysis must answer

1. How much did downtime cost over 18 months, and how does it split between planned and unplanned work?
2. Which assets drive the cost, and how concentrated is it?
3. Which failure types recur, and how do frequency and severity differ?
4. Where is preventive maintenance not happening, and what is that costing?
5. Is restoration time getting worse — by shift, and over the period?

Each question becomes one analysis block in the workbook. Area and criticality are handled as secondary cuts within question 2 rather than as separate questions.

Questions deliberately excluded. Production efficiency and OEE cannot be calculated: the log records downtime but not production output, available calendar hours, or run rates. Any efficiency figure would be invented.

3. Cost model

The source log records spares cost only. Labour cost and production loss are calculated from the client-supplied rates held on the Rates sheet.

Labour_Cost = Labour_Hours × 480 (Rates!B3)

Production_Loss = Downtime_Hours × Loss_Rate[Equipment_Type] if Unplanned
= 0 if Planned

$\text{Total_Event_Cost} = \text{Parts_Cost_INR} + \text{Labour_Cost} + \text{Production_Loss}$

Loss rates per downtime hour, by equipment type: Air Compressor ₹9,500 · Overhead Crane ₹7,200 · Forming Machine ₹6,100 · Cutting Machine ₹5,400 · Surface Treatment ₹3,300 · Welding Equipment ₹2,800.

Excluded from the model: depreciation, plant overheads, idle operator labour standing at a stopped machine, expedited freight on spares, and any warranty or contract recovery. These are excluded because the log carries no data on them, not because they are immaterial.

Held as parameters, not constants. Both rate sets are read from the source file rather than written into formulas. Changing a rate and refreshing updates every figure in the workbook. The client can therefore test their own assumptions without the analyst.

3.1 Verdict on the zero-loss assumption for planned work

The client's stated basis is that planned maintenance is scheduled into non-production windows and therefore carries no production loss.

Accepted. In my own experience of planning fabrication work, a preventive maintenance window is built into the schedule in advance, so the hours it consumes were never counted on for production. Charging production loss against them would double-count time the plan had already released.

One exception to flag: if a planned job overruns its allotted window, the overrun hours do displace production and should be costed. The current log records planned duration but not the window it was allotted, so overruns cannot be identified from this data.

4. Control totals

Established before any analysis, by direct calculation on the raw log. Every figure produced downstream must reconcile to these.

Measure	Value
Events	686
Total downtime hours	3,845.0
— Unplanned	3,579.2 (628 events)
— Planned	265.8 (58 events)
Total labour hours	4,145.2
Parts cost	₹43,36,500
Labour cost	₹19,89,696
Production loss	₹2,36,11,200
Total cost	₹2,99,37,396
— Planned share	₹5,27,414 (1.8%)

5. Assumptions and limitations

1. **Loss-rate sensitivity.** Production loss is ₹2.36 crore of the ₹2.99 crore total (79%). That share is a direct consequence of the client's stated loss rates, which are 6–20× the labour rate. It is arithmetic on an assumption, not an independent finding, and must be validated against actual output data before it drives any investment decision.
 2. **Downtime definition.** It is not established whether Downtime_Hours measures *asset stopped* or *production stopped*. If work continued on other equipment during a stoppage, production loss is overstated.
 3. **Shift comparison.** Shifts are compared on mean hours per event only. Event counts by shift are not reported, because exposure hours per shift are not recorded and counts cannot be interpreted without them.
 4. **Causation.** Where preventive maintenance frequency and breakdown frequency move together, direction cannot be established from this data. Skipped PM may cause breakdowns, or frequent breakdowns may crowd out the PM window. Both are consistent with the evidence, and both point to the same remedy.
 5. **Duty cycle.** Comparisons between paired assets of the same type assume equal duty cycles. If one runs lead and one standby, those comparisons do not hold. This is unconfirmed.
 6. **Forecast.** The projection is a linear trend fitted on 18 monthly observations. Backtested on a 15-month fit against three held-out months, it was 22.5% off on average. Seasonality is not modelled — 18 points is too short to distinguish a seasonal pattern from noise.
 7. **Data completeness.** The final month (Jul-26) was verified complete: the maximum event-end timestamp is 30-Jul-2026 and no events are left open.
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6. Questions for the client

1. Is Downtime_Hours measured as asset stopped or as production stopped?
 2. Is there a recording threshold — are stoppages under 30 minutes logged, or only events raising a work order?
 3. Is the ₹480/hour labour rate a blended internal cost rate or a charge-out rate, and does it already include AMC vendor invoicing?
 4. Do the production-loss rates represent revenue or contribution margin, and do they already include idle operator time?
 5. Is there a documented preventive maintenance schedule with planned-versus-actual compliance, separate from this log?
 6. Are paired assets of the same type duty-cycled equally, or is one lead and one standby?
 7. Is cost data available at project or work-order level, to relate downtime to specific job delays?
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7. Method

Stage	Approach
Shaping	Power Query — explicit types, rates merged from the source file, cost columns calculated in the query rather than on the sheet
Analysis	Five pivot blocks on the Data Model, one per question
Presentation	Interactive dashboard: five KPI cards, five charts, four slicers, one timeline, all connected to a single cache
Forecast	Linear trend with a judgement band, backtested against held-out months
Refresh	Replace the source file, Data → Refresh All. Rates editable on the source Rates sheet

8. Revision note

This is the second version. The first was written before the source file had been read in full, and contained two client questions whose answers were already in the file (criticality definitions and the asset inventory), one question that could not be answered with the columns available (production efficiency), one that assumed shift labels the data does not use, and a cost model stated in prose rather than as a checkable formula. The most important question — where preventive maintenance is not happening and what it costs — was missing entirely.

Recorded here because the correction is part of the method: read the source completely before scoping, and write the model so someone else can reproduce the number without asking.